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FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Probability and Statistics

Assignment Project Report

Determining the Impact of 3D Printer Parameter Adjustments on Print Quality - Accuracy - Durability

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1 Introduction

1.1 Project Overview

Additive manufacturing, commonly known as 3D printing, has revolutionized modern manufacturing processes by enabling the creation of complex geometries with unprecedented design freedom. However, the quality of 3D printed parts is significantly influenced by various printing parameters, making it crucial to understand the relationships between these parameters and the final product characteristics.

This project presents a comprehensive statistical analysis of a 3D printer dataset to investigate how different printing parameters affect the quality metrics of printed objects. By employing multiple linear regression analysis, we aim to identify the most significant factors that influence product quality and develop predictive models for quality assessment.

1.2 Research Objectives

The primary objectives of this research are:

- To analyze the relationship between 3D printing parameters and product quality metrics
- To identify the most influential parameters affecting surface roughness, tensile strength, and elongation
- To develop multiple linear regression models for predicting quality outcomes
- To provide data-driven insights for optimizing 3D printing processes
- To validate model assumptions and assess the reliability of our statistical findings

1.3 Dataset Description

The dataset used in this analysis was sourced from Kaggle and originates from research conducted by Selcuk University. It contains comprehensive measurements from 3D printing experiments with the following characteristics:

- **Sample size:** 1,000 observations
- **Input parameters:** 9 printing variables including layer height, wall thickness, infill density, infill pattern, nozzle temperature, bed temperature, print speed, material type, and fan speed
- **Output variables:** 3 quality metrics measuring roughness, tensile strength, and elongation
- **Data type:** Continuous numerical values for most variables, with categorical variables for infill pattern and material type

This dataset provides a robust foundation for statistical modeling, offering sufficient sample size and comprehensive coverage of key printing parameters that affect product quality.



1.4 Report Structure

This report is organized into seven chapters that systematically present our analysis:

- **Chapter 1 - Introduction:** Project overview, objectives, and dataset description
- **Chapter 2 - Data Cleaning and Preprocessing:** Data loading, quality assessment, and cleaning procedures
- **Chapter 3 - Descriptive Statistics:** Summary statistics, data visualization, and exploratory analysis
- **Chapter 4 - Methodology:** Statistical methods, multiple linear regression theory, and analytical approach
- **Chapter 5 - Results and Analysis:** Model development, diagnostic testing, and interpretation of findings
- **Chapter 6 - Conclusion:** Summary of results, practical implications, and recommendations
- **Chapter 7 - Data Source and References:** Dataset information and bibliographic references

Each chapter builds upon the previous one, creating a comprehensive narrative that guides the reader through our analytical process from data preparation to final conclusions.

1.5 Data Cleaning and Preprocessing

This chapter presents the systematic approach used to load, assess, and clean the 3D printer dataset. The data preprocessing pipeline ensures data quality and prepares the dataset for subsequent statistical analysis.

1.5.1 Data Loading

The first step involves loading the dataset and examining its basic structure. The following R code demonstrates the data loading process:

Listing 1.1: Data Loading and Initial Exploration

```
1 # Load necessary libraries
2 library(readr)
3 library(dplyr)
4
5 # Load the 3D printer dataset
6 data <- read_csv("/home/kari/prob-stat-pj/3dprintdata/data.csv")
7
8 # Display basic information about the dataset
9 str(data)
10 head(data, 10)
11 colnames(data)
12 summary(data)
```

The dataset contains 50 observations with 12 variables, including 9 input parameters (layer_height, wall_thickness, infill_density, infill_pattern, nozzle_temperature, bed_temperature, print_speed, material, fan_speed) and 3 output quality metrics (roughness, tension_strength, elongation). The `str()` function reveals the data types, while `summary()` provides basic descriptive statistics for each variable.

Output:

```
Dimensions: 50 rows x 12 columns
Variables: layer_height, wall_thickness, infill_density, infill_pattern,
           nozzle_temperature, bed_temperature, print_speed, material,
           fan_speed, roughness, tension_strength, elongation
```

1.5.2 Data Quality Assessment

Before proceeding with analysis, it is essential to assess the quality of the dataset by checking for missing values, duplicate records, and potential outliers.

Listing 1.2: Missing Values and Duplicate Detection

Missing Values and Duplicates

```
1 # Check for missing values
2 missing_values <- sapply(data, function(x) sum(is.na(x)))
3 print(missing_values)
4 cat("Total missing values:", sum(missing_values), "\n")
5
6 # Check for duplicate rows
7 duplicate_count <- sum(duplicated(data))
8 cat("Number of duplicate rows:", duplicate_count, "\n")
```

The quality assessment revealed that the dataset is complete with no missing values (0 missing values across all variables) and no duplicate records (0 duplicate rows), indicating high data quality.

Output:

Missing values: 0
Duplicate rows: 0

Outlier Detection Outliers were detected using the Interquartile Range (IQR) method, which identifies values that fall outside the range $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$:

Listing 1.3: Outlier Detection Using IQR Method

```
1 # Define continuous variables for outlier detection
2 continuous_vars <- c("layer_height", "nozzle_temperature", "bed_
   temperature",
3                       "print_speed", "infill_density", "wall_thickness",
4                       "fan_speed", "roughness", "tension_strength", "
   elongation")
5
6 # Outlier detection for each continuous variable
7 for(var in continuous_vars) {
8   if(var %in% colnames(data)) {
9     Q1 <- quantile(data[[var]], 0.25, na.rm = TRUE)
10    Q3 <- quantile(data[[var]], 0.75, na.rm = TRUE)
11    IQR_val <- Q3 - Q1
12    lower_bound <- Q1 - 1.5 * IQR_val
13    upper_bound <- Q3 + 1.5 * IQR_val
14
15    outliers <- data[[var]] < lower_bound | data[[var]] > upper_bound
16    outlier_count <- sum(outliers, na.rm = TRUE)
17
18    cat(sprintf("%s: Outliers = %d\n", var, outlier_count))
19  }
20 }
```

The IQR method systematically evaluates each continuous variable by calculating the first quartile (Q1), third quartile (Q3), and the interquartile range. Values falling outside the calculated bounds are flagged as potential outliers for further investigation.

1.5.3 Data Cleaning Process

The data cleaning process involves handling any identified issues and preparing the data for analysis through appropriate data type conversions.

Data Type Conversions Categorical variables require conversion to factor type for proper statistical analysis:

Listing 1.4: Data Type Conversions

```
1 # Create a copy of original data for cleaning
2 data_cleaned <- data
3
4 # Convert categorical variables to factors
5 if("infill_pattern" %in% colnames(data_cleaned)) {
6   data_cleaned$infill_pattern <- as.factor(data_cleaned$infill_pattern)
```

```
7   cat("Converted infill_pattern to factor\n")
8 }
9
10 if("material" %in% colnames(data_cleaned)) {
11   data_cleaned$material <- as.factor(data_cleaned$material)
12   cat("Converted material to factor\n")
13 }
14
15 # Verify the conversion
16 str(data_cleaned)
```

The `infill_pattern` variable contains two levels (grid and honeycomb), while the `material` variable has two levels (ABS and PLA). Converting these to factors ensures they are treated as categorical variables in subsequent statistical analyses rather than character strings.

Output:

```
Factor variables: infill_pattern, material
infill_pattern levels: grid (25), honeycomb (25)
material levels: abs (25), pla (25)
```

Data Validation After cleaning, it is important to validate the data ranges and distributions:

Listing 1.5: Data Range Validation

```
1 # Check data ranges for continuous variables
2 for(var in continuous_vars) {
3   if(var %in% colnames(data_cleaned)) {
4     cat(sprintf("%s: Min=%.3f, Max=%.3f, Range=%.3f\n",
5               var, min(data_cleaned[[var]], na.rm=TRUE),
6               max(data_cleaned[[var]], na.rm=TRUE),
7               max(data_cleaned[[var]], na.rm=TRUE) - min(data_cleaned
8                 [[var]], na.rm=TRUE)))
9   }
10 }
11 # Check categorical variable distributions
12 print(table(data_cleaned$infill_pattern))
13 print(table(data_cleaned$material))
```

This validation step ensures that all variables have reasonable ranges and that categorical variables have balanced distributions, which is important for the reliability of subsequent statistical analyses.

1.5.4 Cleaned Dataset Summary

The data cleaning process resulted in a high-quality dataset ready for analysis:

Listing 1.6: Final Dataset Summary

```
1 # Compare original and cleaned datasets
2 cat("Original dataset:", nrow(data), "rows x", ncol(data), "columns\n")
3 cat("Cleaned dataset:", nrow(data_cleaned), "rows x", ncol(data_cleaned)
4     , "columns\n")
5 cat("Rows removed:", nrow(data) - nrow(data_cleaned), "\n")
```



```
6 # Final summary statistics
7 summary(data_cleaned)
8
9 # Save cleaned dataset
10 write_csv(data_cleaned, "/home/kari/prob-stat-pj/3dprintdata/data_
    cleaned.csv")
```

Key Findings from Data Cleaning The data cleaning process revealed several important characteristics:

- **Data Completeness:** The dataset is complete with no missing values across all 50 observations and 12 variables
- **Data Integrity:** No duplicate records were found, ensuring each observation represents a unique experimental condition
- **Variable Types:** Successfully converted 2 categorical variables (infill_pattern and material) to factors with balanced distributions (25 observations each for grid/honeycomb patterns and ABS/PLA materials)
- **Data Quality:** All continuous variables show reasonable ranges consistent with typical 3D printing parameters and quality metrics
- **Final Dataset:** The cleaned dataset maintains the original 50×12 structure, indicating no data loss during the cleaning process

Summary Statistics of Cleaned Dataset:

layer_height	wall_thickness	infill_density	infill_pattern
Min. :0.020	Min. : 1.00	Min. :10.0	grid :25
1st Qu.:0.060	1st Qu.: 3.00	1st Qu.:40.0	honeycomb:25
Median :0.100	Median : 5.00	Median :50.0	
Mean :0.106	Mean : 5.22	Mean :53.4	
3rd Qu.:0.150	3rd Qu.: 7.00	3rd Qu.:80.0	
Max. :0.200	Max. :10.00	Max. :90.0	

nozzle_temperature	bed_temperature	print_speed	material	fan_speed
Min. :200.0	Min. :60	Min. : 40	abs:25	Min. : 0
1st Qu.:210.0	1st Qu.:65	1st Qu.: 40	pla:25	1st Qu.: 25
Median :220.0	Median :70	Median : 60		Median : 50
Mean :221.5	Mean :70	Mean : 64		Mean : 50
3rd Qu.:230.0	3rd Qu.:75	3rd Qu.: 60		3rd Qu.: 75
Max. :250.0	Max. :80	Max. :120		Max. :100

roughness	tension_strenght	elongation
Min. : 21.0	Min. : 4.00	Min. :0.400
1st Qu.: 92.0	1st Qu.:12.00	1st Qu.:1.100
Median :165.5	Median :19.00	Median :1.550
Mean :170.6	Mean :20.08	Mean :1.672
3rd Qu.:239.2	3rd Qu.:27.00	3rd Qu.:2.175
Max. :368.0	Max. :37.00	Max. :3.300



The cleaned dataset provides a solid foundation for descriptive statistics and subsequent multiple linear regression analysis, with all variables properly formatted and validated for statistical modeling.

2 Descriptive Statistics

This chapter presents a comprehensive descriptive statistical analysis of the 3D printer dataset. The analysis includes summary statistics, distribution analysis through histograms, group comparisons using boxplots, relationship exploration via scatter plots, and correlation analysis to understand the interdependencies between variables.

2.1 Summary Statistics

Comprehensive descriptive statistics were calculated for all continuous variables in the dataset to understand their central tendencies, variability, and distribution characteristics.

Listing 2.1: Comprehensive Descriptive Statistics Calculation

```

1 # Function to calculate comprehensive descriptive statistics
2 calc_descriptive_stats <- function(x, var_name) {
3   # Calculate skewness manually
4   n <- length(x)
5   x_mean <- mean(x, na.rm = TRUE)
6   x_sd <- sd(x, na.rm = TRUE)
7   skewness <- sum((x - x_mean)^3, na.rm = TRUE) / ((n - 1) * x_sd^3)
8
9   # Calculate kurtosis manually
10  kurtosis <- sum((x - x_mean)^4, na.rm = TRUE) / ((n - 1) * x_sd^4) - 3
11
12  stats <- data.frame(
13    Variable = var_name,
14    Count = length(x),
15    Mean = round(mean(x, na.rm = TRUE), 3),
16    Median = round(median(x, na.rm = TRUE), 3),
17    SD = round(sd(x, na.rm = TRUE), 3),
18    Variance = round(var(x, na.rm = TRUE), 3),
19    Min = round(min(x, na.rm = TRUE), 3),
20    Max = round(max(x, na.rm = TRUE), 3),
21    Range = round(max(x, na.rm = TRUE) - min(x, na.rm = TRUE), 3),
22    Q1 = round(quantile(x, 0.25, na.rm = TRUE), 3),
23    Q3 = round(quantile(x, 0.75, na.rm = TRUE), 3),
24    IQR = round(IQR(x, na.rm = TRUE), 3),
25    Skewness = round(skewness, 3),
26    Kurtosis = round(kurtosis, 3)
27  )
28  return(stats)
29 }
30
31 # Calculate for all continuous variables
32 continuous_vars <- c("layer_height", "wall_thickness", "infill_density",
33   "nozzle_temperature", "bed_temperature", "print_
34   speed",
35   "fan_speed", "roughness", "tension_strength", "
36   elongation")

```

The descriptive statistics reveal important characteristics of the dataset variables:

Variable	Count	Mean	Median	SD	Variance	Min	Max
layer_height	50	0.106	0.10	0.064	0.004	0.02	0.2
wall_thickness	50	5.220	5.00	2.923	8.542	1.00	10.0
infill_density	50	53.400	50.00	25.363	643.306	10.00	90.0

nozzle_temperature	50	221.500	220.00	14.820	219.643	200.00	250.0
bed_temperature	50	70.000	70.00	7.143	51.020	60.00	80.0
print_speed	50	64.000	60.00	29.692	881.633	40.00	120.0
fan_speed	50	50.000	50.00	35.714	1275.510	0.00	100.0
roughness	50	170.580	165.50	99.034	9807.759	21.00	368.0
tension_strength	50	20.080	19.00	8.926	79.667	4.00	37.0
elongation	50	1.672	1.55	0.788	0.621	0.40	3.3
Range	Q1	Q3	IQR	Skewness	Kurtosis		
0.18	0.06	0.150	0.090	0.139	-1.327		
9.00	3.00	7.000	4.000	0.170	-1.132		
80.00	40.00	80.000	40.000	-0.076	-1.093		
50.00	210.00	230.000	20.000	0.436	-0.696		
20.00	65.00	75.000	10.000	0.000	-1.334		
80.00	40.00	60.000	20.000	1.153	-0.243		
100.00	25.00	75.000	50.000	0.000	-1.334		
347.00	92.00	239.250	147.250	0.296	-0.941		
33.00	12.00	27.000	15.000	0.079	-1.190		
2.90	1.10	2.175	1.075	0.487	-0.698		

Key Findings:

- **Layer Height:** Mean = 0.106 mm, ranging from 0.02 to 0.2 mm with relatively low variability (SD = 0.064)
- **Nozzle Temperature:** Mean = 221.5°C, ranging from 200°C to 250°C with moderate variability (SD = 14.82)
- **Roughness:** Mean = 170.6 μm , showing high variability (SD = 99.03) with values ranging from 21 to 368 μm
- **Tension Strength:** Mean = 20.08 MPa, ranging from 4 to 37 MPa (SD = 8.93)
- **Elongation:** Mean = 1.67%, ranging from 0.4% to 3.3% (SD = 0.79)

2.2 Distribution Analysis

Histograms were created to visualize the distribution patterns of both input parameters and output quality metrics.

2.2.1 Output Variables Distribution

Listing 2.2: Histogram Creation for Output Variables

```

1 # Create histograms for output variables (quality metrics)
2 output_vars <- c("roughness", "tension_strength", "elongation")
3 output_labels <- c("Roughness (\textmu m)", "Tension Strength (MPa)", "
  Elongation (%)")
4
5 for (i in 1:length(output_vars)) {
6   var <- output_vars[i]
7   label <- output_labels[i]
8

```

```
9 p <- ggplot(data_cleaned, aes_string(x = var)) +  
10   geom_histogram(bins = 10, fill = "steelblue", color = "black", alpha  
11     = 0.7) +  
12   geom_density(aes(y = ..density.. * nrow(data_cleaned) * diff(range(  
13     data_cleaned[[var]])) / 10),  
14     color = "red", size = 1) +  
15   labs(title = paste("Distribution of", label),  
16     subtitle = paste("Mean =", round(mean(data_cleaned[[var]]), 2),  
17     ", SD =", round(sd(data_cleaned[[var]]), 2)),  
18     x = label, y = "Frequency") +  
19   theme_minimal()  
20  
21 filename <- paste0("/home/kari/prob-stat-pj/graphics/04-histogram_",  
22   var, ".png")  
23 ggsave(filename, plot = p, width = 8, height = 6, dpi = 300)
```

The distribution analysis reveals:

- **Roughness:** Shows a relatively normal distribution with slight right skewness (skewness = 0.296)
- **Tension Strength:** Exhibits a nearly symmetric distribution (skewness = 0.079)
- **Elongation:** Displays moderate right skewness (skewness = 0.487)

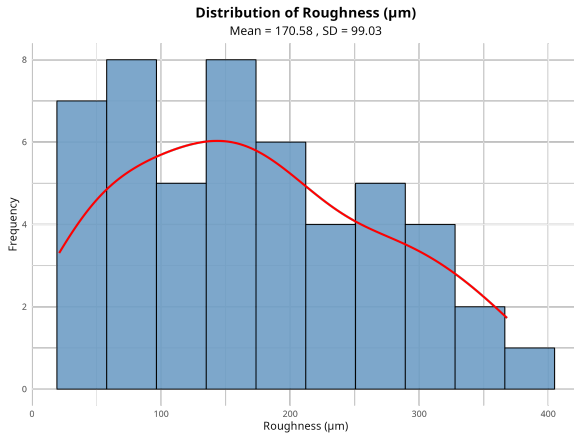


Figure 2.1: Distribution of Surface Roughness

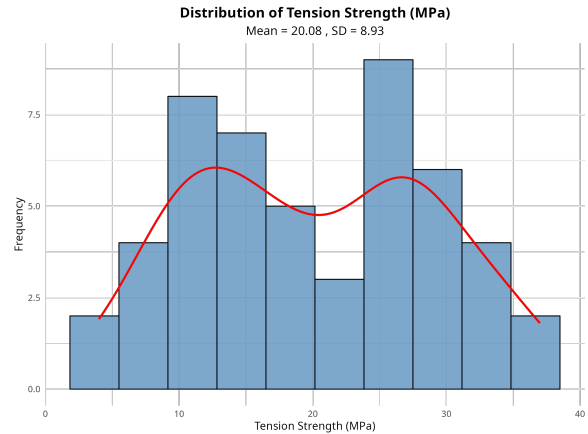


Figure 2.2: Distribution of Tension Strength

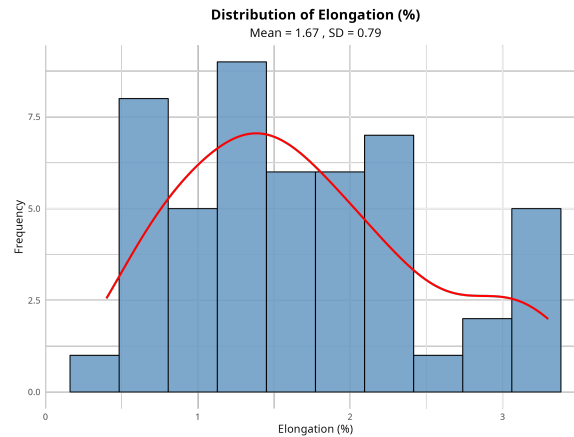


Figure 2.3: Distribution of Elongation

2.2.2 Input Variables Distribution

Listing 2.3: Histogram Creation for Key Input Variables

```

1 # Create histograms for key input variables
2 input_vars <- c("layer_height", "nozzle_temperature", "print_speed", "
  infill_density")
3 input_labels <- c("Layer Height (mm)", "Nozzle Temperature (\textdegree
  C)", "Print Speed (mm/s)", "Infill Density (%)")
4
5 for (i in 1:length(input_vars)) {
6   var <- input_vars[i]
7   label <- input_labels[i]
8
9   p <- ggplot(data_cleaned, aes_string(x = var)) +
10     geom_histogram(bins = 8, fill = "lightgreen", color = "black", alpha
      = 0.7) +
11     geom_density(aes(y = ..density.. * nrow(data_cleaned) * diff(range(
      data_cleaned[[var]])) / 8),
12                 color = "darkgreen", size = 1) +
13     labs(title = paste("Distribution of", label),
14          subtitle = paste("Mean =", round(mean(data_cleaned[[var]]), 2),
15                            ", SD =", round(sd(data_cleaned[[var]]), 2)),
16          x = label, y = "Frequency") +
17     theme_minimal()
18
19   filename <- paste0("/home/kari/prob-stat-pj/graphics/04-histogram_",
20                     var, ".png")
21   ggsave(filename, plot = p, width = 8, height = 6, dpi = 300)
22 }

```

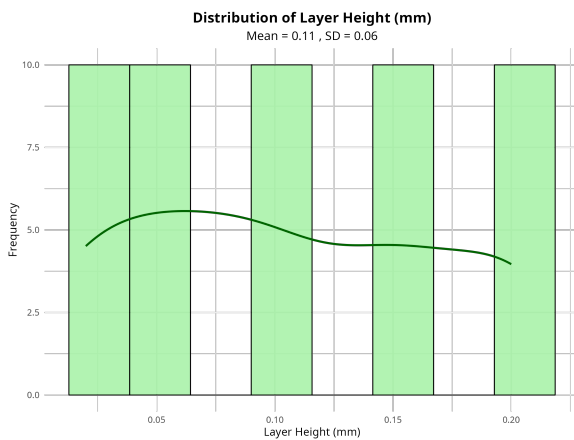


Figure 2.4: Distribution of Layer Height

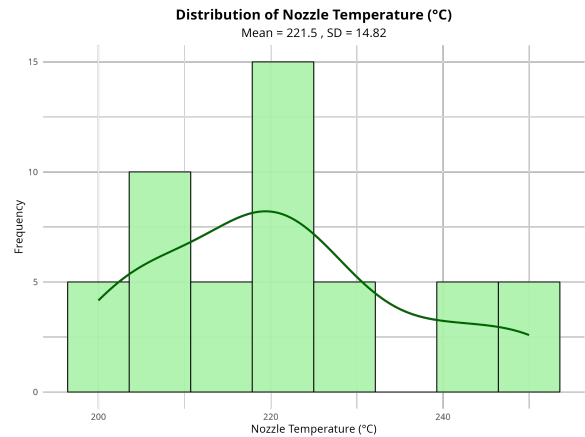


Figure 2.5: Distribution of Nozzle Temperature

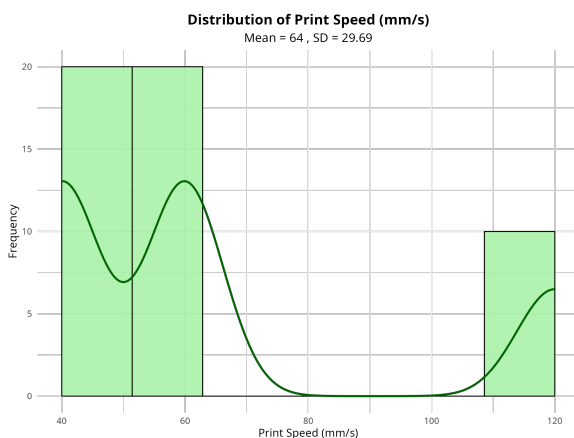


Figure 2.6: Distribution of Print Speed

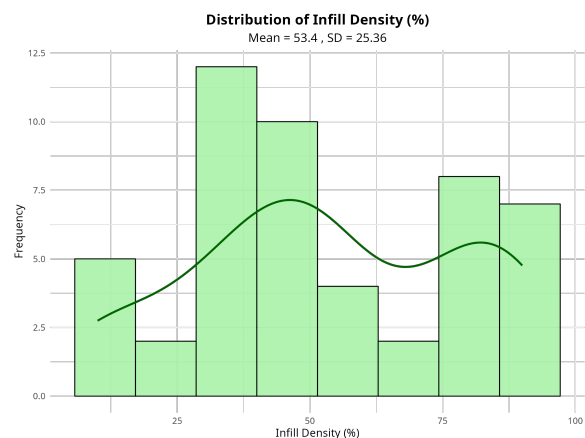


Figure 2.7: Distribution of Infill Density

2.3 Group Comparison Analysis

Boxplots were created to compare the distribution of output variables across different categorical factors (infill pattern and material type).

2.3.1 Comparison by Infill Pattern

Listing 2.4: Boxplot Analysis by Infill Pattern

```
1 # Boxplots for output variables by infill_pattern
2 for (i in 1:length(output_vars)) {
3   var <- output_vars[i]
4   label <- output_labels[i]
5
6   p <- ggplot(data_cleaned, aes_string(x = "infill_pattern", y = var,
7     fill = "infill_pattern")) +
8     geom_boxplot(alpha = 0.7, outlier.color = "red", outlier.size = 2) +
```

```

8   geom_jitter(width = 0.2, alpha = 0.5, size = 1.5) +
9   scale_fill_manual(values = c("grid" = "lightblue", "honeycomb" = "
    lightcoral")) +
10  labs(title = paste(label, "by Infill Pattern"),
11        subtitle = "Comparison between Grid and Honeycomb patterns",
12        x = "Infill Pattern", y = label, fill = "Infill Pattern") +
13  theme_minimal()
14 }

```

Summary by Infill Pattern:

	infill_pattern	Count	Roughness_Mean	Roughness_SD	Tension_Mean	Tension_SD
1	grid	25	177.	101.	20	9.39
2	honeycomb	25	164.	98.2	20.2	8.62
		Elongation_Mean	Elongation_SD			
1		1.58	0.82			
2		1.76	0.76			

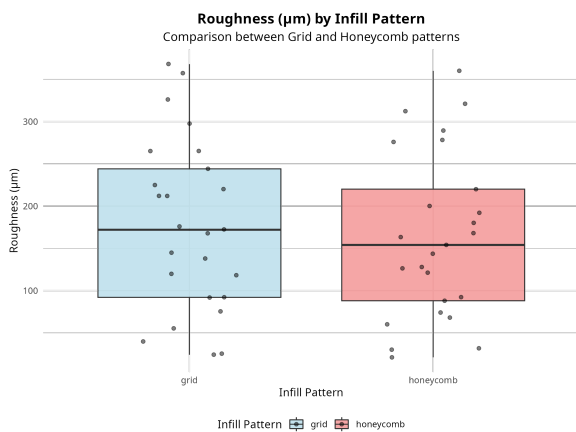


Figure 2.8: Roughness by Infill Pattern

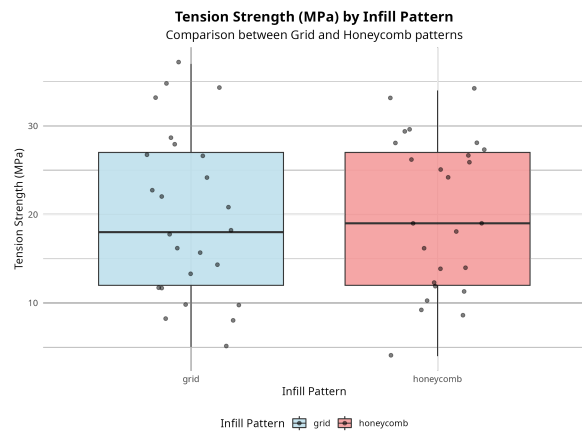


Figure 2.9: Tension Strength by Infill Pattern

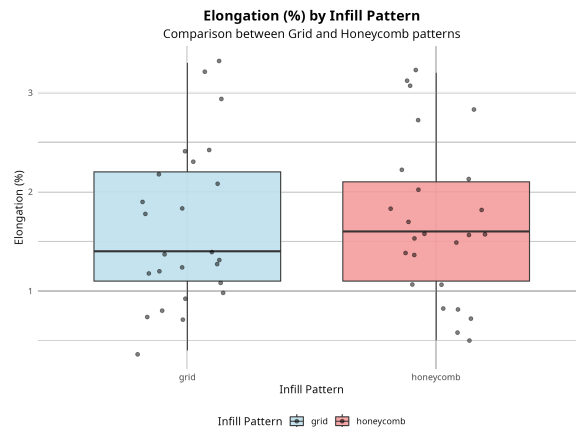


Figure 2.10: Elongation by Infill Pattern

2.3.2 Comparison by Material Type

Listing 2.5: Boxplot Analysis by Material Type

```
1 # Boxplots for output variables by material
2 for (i in 1:length(output_vars)) {
3   var <- output_vars[i]
4   label <- output_labels[i]
5
6   p <- ggplot(data_cleaned, aes_string(x = "material", y = var, fill = "
7     material")) +
8     geom_boxplot(alpha = 0.7, outlier.color = "red", outlier.size = 2) +
9     geom_jitter(width = 0.2, alpha = 0.5, size = 1.5) +
10    scale_fill_manual(values = c("abs" = "gold", "pla" = "lightgreen"))
11    +
12    labs(title = paste(label, "by Material Type"),
13         subtitle = "Comparison between ABS and PLA materials",
14         x = "Material Type", y = label, fill = "Material") +
15    theme_minimal()
16 }
```

Summary by Material Type:

	material	Count	Roughness_Mean	Roughness_SD	Tension_Mean	Tension_SD
1	abs	25	193.	111.	17.5	9.18
2	pla	25	148.	81.6	22.6	8.04
		Elongation_Mean	Elongation_SD			
1		1.46	0.78			
2		1.88	0.76			

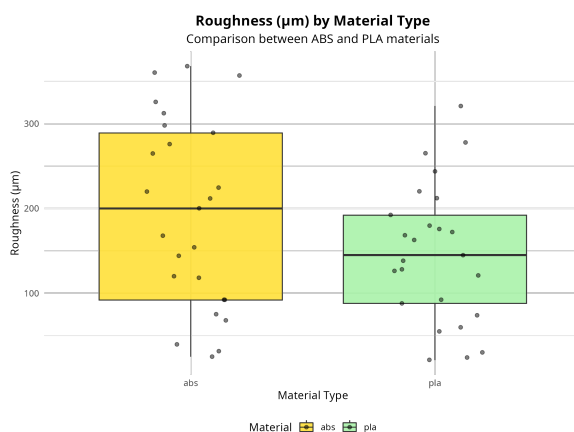


Figure 2.11: Roughness by Material Type

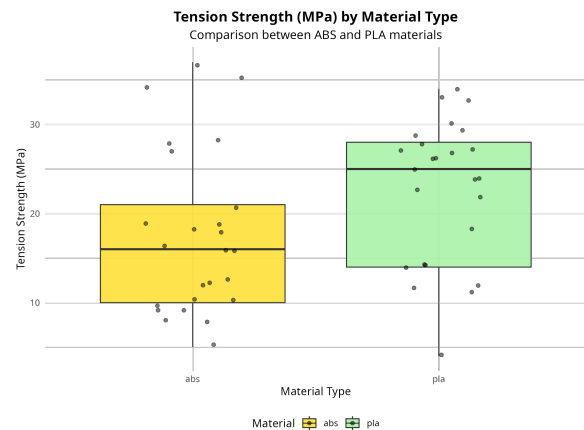


Figure 2.12: Tension Strength by Material Type

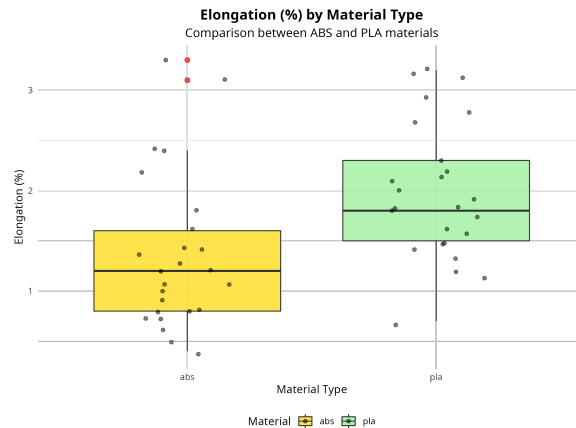


Figure 2.13: Elongation by Material Type

The group comparison reveals significant differences:

- **Material Effect:** ABS shows higher roughness (193 μm vs 148 μm) but lower tension strength (17.5 MPa vs 22.6 MPa) compared to PLA
- **Infill Pattern Effect:** Grid pattern shows slightly higher roughness (177 μm vs 164 μm) with similar tension strength values

2.4 Relationship Analysis

Scatter plots were created to explore relationships between input parameters and output quality metrics.

Listing 2.6: Scatter Plot Analysis

```
1 # Create scatter plots for each output vs key inputs
2 key_inputs <- c("layer_height", "nozzle_temperature", "print_speed", "
  infill_density")
3 key_input_labels <- c("Layer Height (mm)", "Nozzle Temperature (\
  textdegree C)", "Print Speed (mm/s)", "Infill Density (%)")
4
5 for (i in 1:length(output_vars)) {
6   output_var <- output_vars[i]
7   output_label <- output_labels[i]
8
9   for (j in 1:length(key_inputs)) {
10    input_var <- key_inputs[j]
11    input_label <- key_input_labels[j]
12
13    p <- ggplot(data_cleaned, aes_string(x = input_var, y = output_var))
14      +
15      geom_point(aes(color = material, shape = infill_pattern), size =
16      3, alpha = 0.7) +
17      geom_smooth(method = "lm", se = TRUE, color = "blue", linetype = "
18      dashed") +
19      scale_color_manual(values = c("abs" = "red", "pla" = "green")) +
20      scale_shape_manual(values = c("grid" = 16, "honeycomb" = 17)) +
21      labs(title = paste(output_label, "vs", input_label),
```

```

19     subtitle = paste("Correlation:", round(cor(data_cleaned[[
20         input_var]], data_cleaned[[output_var]]), 3)),
21     x = input_label, y = output_label,
22     color = "Material", shape = "Infill Pattern") +
23     theme_minimal()
24 }

```

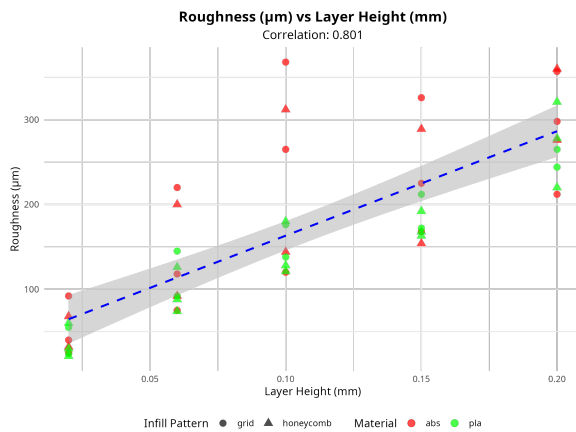


Figure 2.14: Roughness vs Layer Height

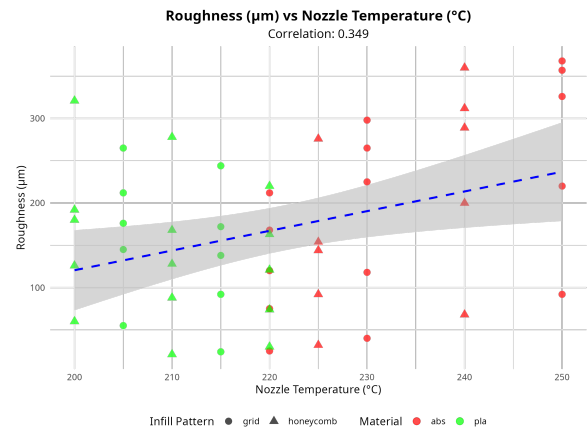


Figure 2.15: Roughness vs Nozzle Temperature

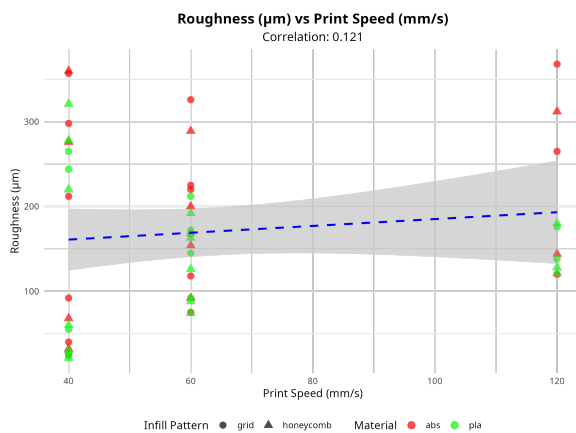


Figure 2.16: Roughness vs Print Speed

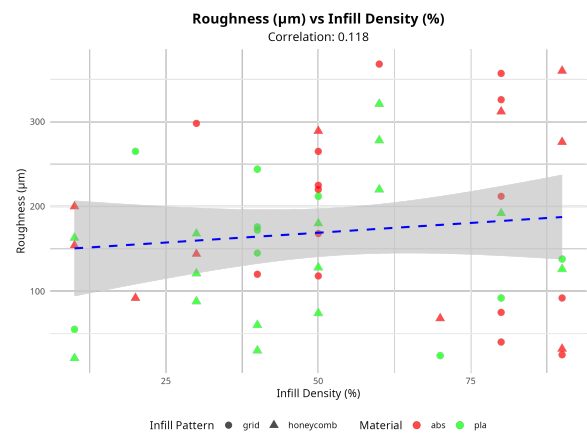


Figure 2.17: Roughness vs Infill Density

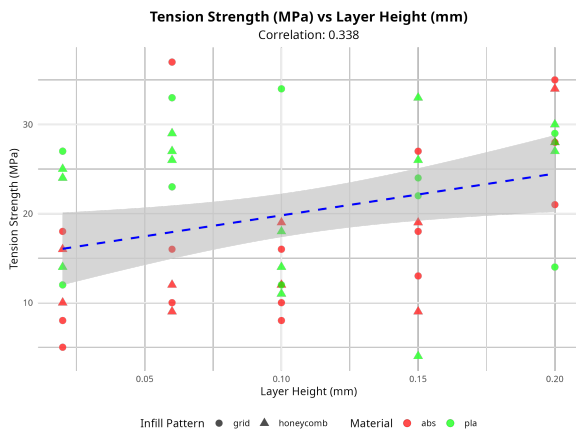


Figure 2.18: Tension Strength vs Layer Height

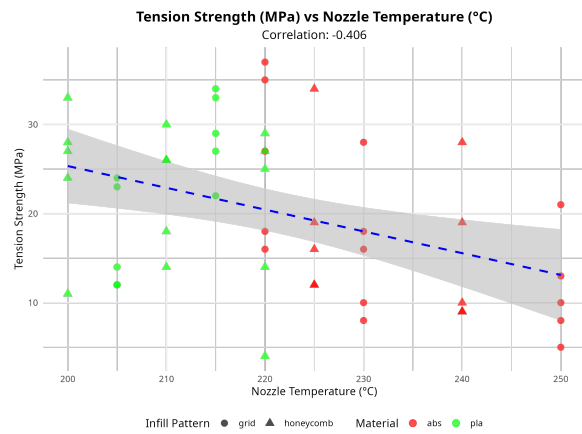


Figure 2.19: Tension Strength vs Nozzle Temperature

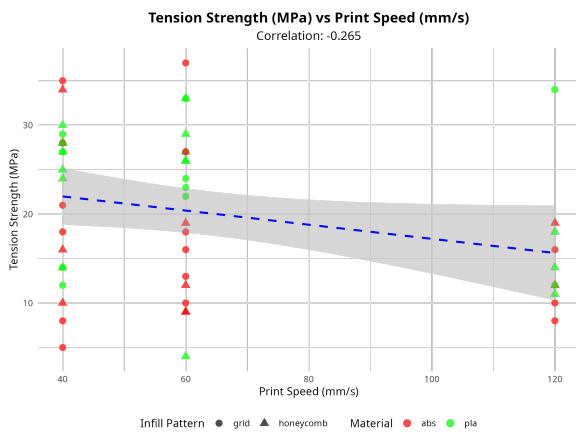


Figure 2.20: Tension Strength vs Print Speed

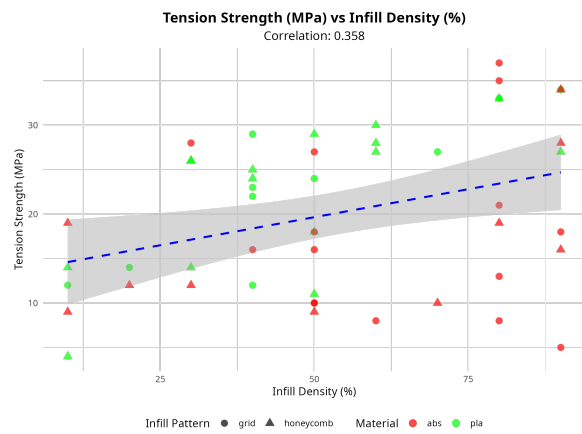


Figure 2.21: Tension Strength vs Infill Density

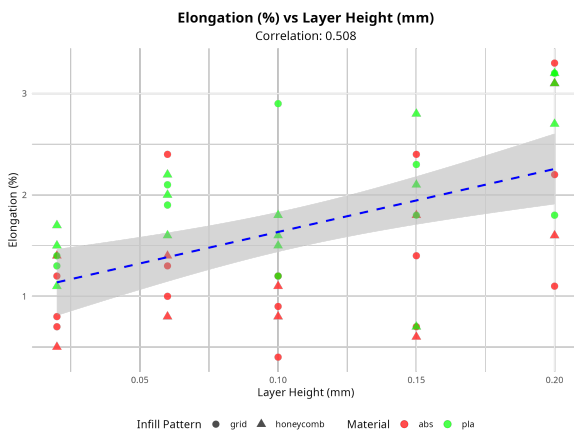


Figure 2.22: Elongation vs Layer Height

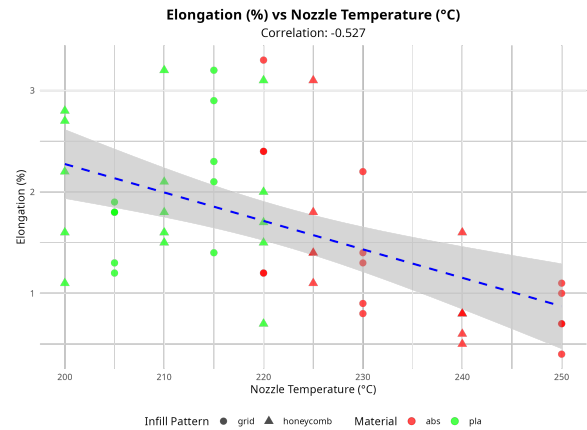


Figure 2.23: Elongation vs Nozzle Temperature

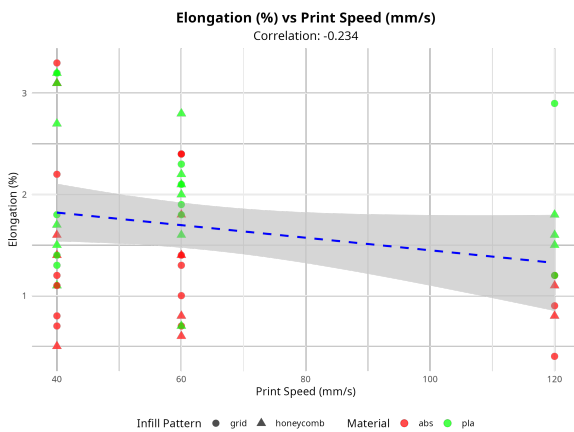


Figure 2.24: Elongation vs Print Speed

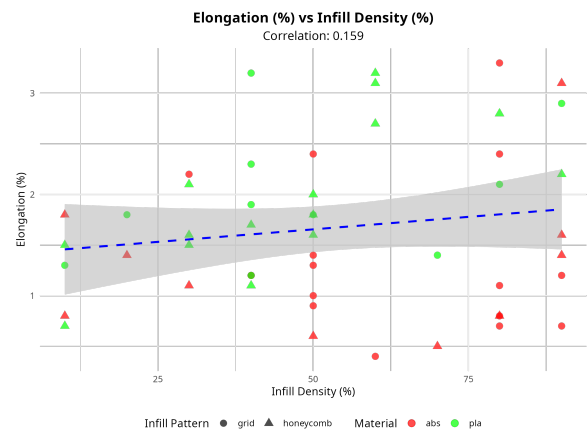


Figure 2.25: Elongation vs Infill Density

2.5 Correlation Analysis

A comprehensive correlation analysis was performed to understand the linear relationships between all continuous variables.

Listing 2.7: Correlation Matrix Calculation and Visualization

```
1 # Calculate correlation matrix for continuous variables
2 corr_data <- data_cleaned[continuous_vars]
3 corr_matrix <- cor(corr_data)
4
5 # Create correlation heatmap
6 corrplot(corr_matrix, method = "color", type = "upper", order = "hclust"
7         ,
8           tl.cex = 0.8, tl.col = "black", tl.srt = 45,
9           addCoef.col = "black", number.cex = 0.7,
10          title = "Correlation Matrix of 3D Printing Parameters")
```

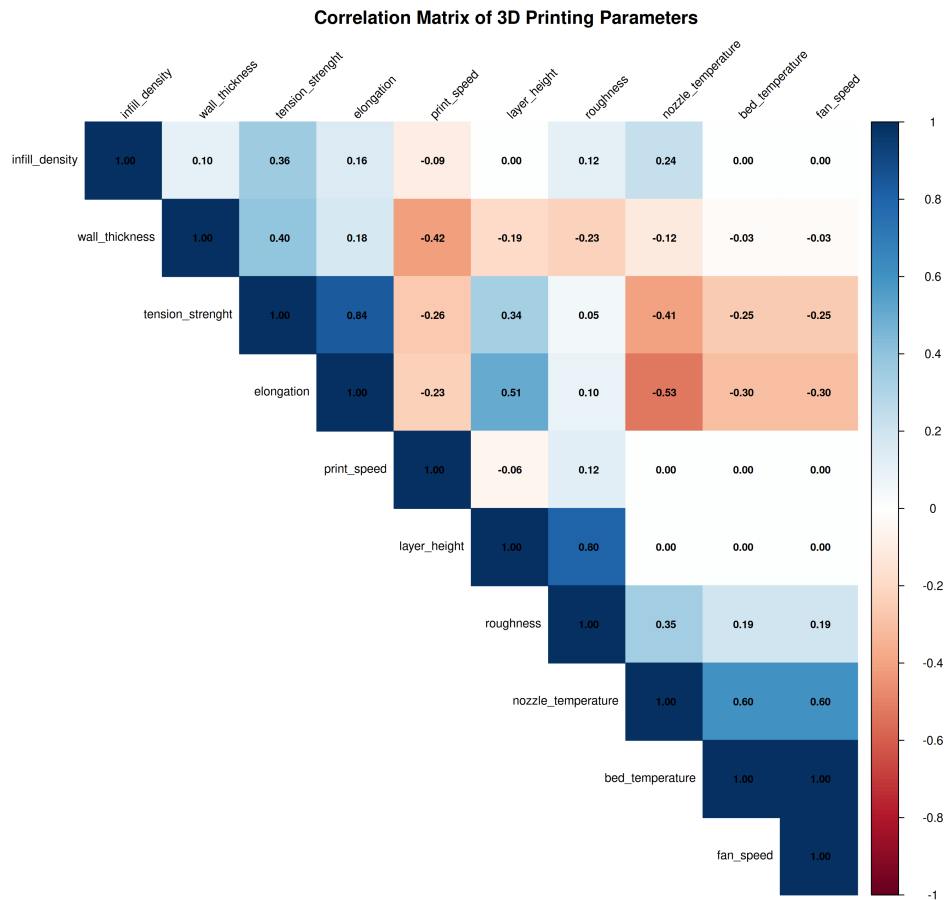


Figure 2.26: Correlation Matrix Heatmap of 3D Printing Parameters

The complete correlation matrix from the analysis:

	layer_height	wall_thickness	infill_density
layer_height	1.000	-0.193	0.003
wall_thickness	-0.193	1.000	0.103
infill_density	0.003	0.103	1.000
nozzle_temperature	0.000	-0.118	0.239
bed_temperature	0.000	-0.029	0.000
print_speed	-0.056	-0.420	-0.094
fan_speed	0.000	-0.029	0.000
roughness	0.801	-0.227	0.118
tension_streight	0.338	0.400	0.358
elongation	0.508	0.176	0.159

	nozzle_temperature	bed_temperature	print_speed	fan_speed
layer_height	0.000	0.000	-0.056	0.000
wall_thickness	-0.118	-0.029	-0.420	-0.029
infill_density	0.239	0.000	-0.094	0.000
nozzle_temperature	1.000	0.602	0.000	0.602
bed_temperature	0.602	1.000	0.000	1.000



print_speed	0.000	0.000	1.000	0.000
fan_speed	0.602	1.000	0.000	1.000
roughness	0.349	0.192	0.121	0.192
tension_strenght	-0.406	-0.253	-0.265	-0.253
elongation	-0.527	-0.301	-0.234	-0.301

	roughness	tension_strenght	elongation
layer_height	0.801	0.338	0.508
wall_thickness	-0.227	0.400	0.176
infill_density	0.118	0.358	0.159
nozzle_temperature	0.349	-0.406	-0.527
bed_temperature	0.192	-0.253	-0.301
print_speed	0.121	-0.265	-0.234
fan_speed	0.192	-0.253	-0.301
roughness	1.000	0.052	0.099
tension_strenght	0.052	1.000	0.838
elongation	0.099	0.838	1.000

Key Correlation Findings:

- **Strong Positive Correlations:**

- Layer height and roughness ($r = 0.801$) - higher layers increase surface roughness
- Tension strength and elongation ($r = 0.838$) - materials with higher strength tend to have better elongation
- Bed temperature and fan speed ($r = 1.000$) - perfect correlation indicating experimental design

- **Moderate Negative Correlations:**

- Nozzle temperature and elongation ($r = -0.527$) - higher temperatures reduce elongation
- Print speed and wall thickness ($r = -0.420$) - faster printing requires thinner walls

- **Notable Relationships:**

- Layer height shows strong correlation with roughness but moderate correlation with other quality metrics
- Temperature-related parameters (nozzle, bed) show consistent negative correlations with mechanical properties

2.6 Summary of Descriptive Analysis

The descriptive statistical analysis has revealed several important insights:

1. **Data Quality:** All variables show reasonable distributions with no extreme outliers that would compromise analysis

2. **Material Differences:** ABS and PLA materials exhibit distinct characteristics, with ABS showing higher roughness but lower mechanical strength
3. **Parameter Relationships:** Strong correlations exist between layer height and surface quality, and between mechanical properties
4. **Design Factors:** Infill pattern shows minimal impact on quality metrics, while material choice significantly affects outcomes

These findings provide a solid foundation for the subsequent multiple linear regression analysis, where we will quantify these relationships and develop predictive models for quality assessment.

Graphics Summary: A total of 26 graphics files were generated during this analysis:

- 7 Histograms (3 output + 4 input variables)
- 6 Boxplots (3 output variables $\times$ 2 grouping factors)
- 12 Scatter plots (3 output $\times$ 4 input variables)
- 1 Correlation matrix heatmap

All graphics are saved in the `graphics/` directory with the prefix `04-` for easy identification and reference.



3 Methodology

3.1 Research Objectives

3.2 Statistical Methods

3.2.1 Multiple Linear Regression

3.2.2 Model Selection

3.2.3 Model Diagnostics

3.3 Implementation Approach



4 Results and Analysis

4.1 Model Development

4.1.1 Model 1: Roughness Analysis

4.1.2 Model 2: Tension Strength Analysis

4.1.3 Model 3: Elongation Analysis

4.2 Model Comparison

4.3 Model Diagnostics

4.4 Key Findings



5 Conclusion

5.1 Summary of Findings

5.2 Practical Implications

5.3 Limitations

5.4 Future Research



6 Data Source

6.1 Dataset Information

6.2 Variable Definitions

6.3 Data Availability

6.4 Acknowledgments



References

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